Parenthood Timing and Gender Inequality

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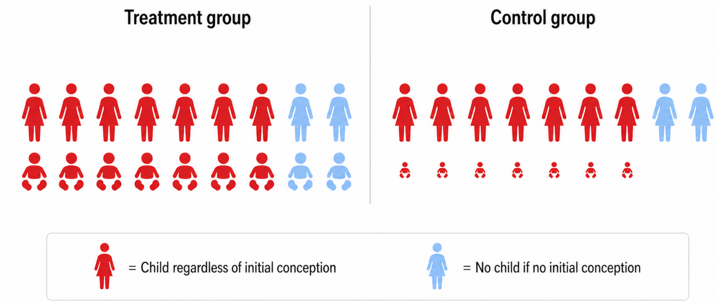
Motivation

How does parenthood affect gender gaps in the labor market?

- A key challenge in answering this question is selection

Existing work uses various quasi-experiments to address selection (IVF, LARC):

- At a fixed moment, some women conceive (treatment), others do not (control)



Differences in career stage at birth, parenthood duration, and perhaps # of children →

- Comparison mixes motherhood vs. childlessness with earlier vs. later motherhood

This Paper

I provide first evidence on the independent effects of parenthood and its timing.

- Focus on couples undergoing intrauterine insemination (IUI)
- New methodology that exploits variation in the full sequence of IUI attempts

Imagine two naive comparisons:

- Parenthood: first-IUI pregnancies vs. childlessness after IUI sequence failure
- Parenthood timing: first-IUI pregnancies vs. pregnancy after IUI sequence failure

Both comparisons are subject to selection:

1. Selection into additional IUI attempts
2. Selection into natural conception after IUI sequence failure

- I show information on # of IUI attempts is sufficient to correct for 1.
- I use assumption-free worse-case bounds to correct for 2.

Only identifying assumption:

- Attempt n success is as-good-as-random **among those who undergo attempt** n.

Simple World: Two Possible IUI Attempts and No Natural Conception

Stops after
first failure

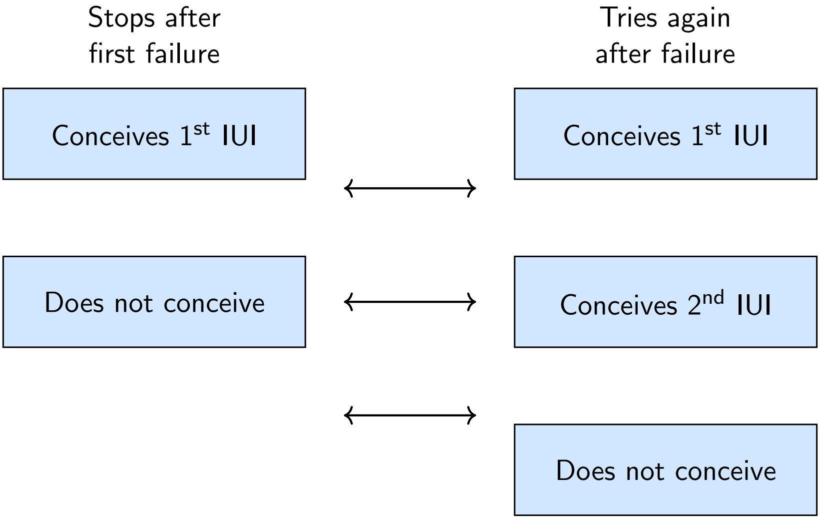
Conceives 1st IUI

Does not conceive

Simple World: Two Possible IUI Attempts and No Natural Conception

Stops after first failure	Tries again after failure
Conceives 1st IUI	Conceives 1st IUI
Does not conceive	Conceives 2nd IUI
	Does not conceive

Simple World: Two Possible IUI Attempts and No Natural Conception



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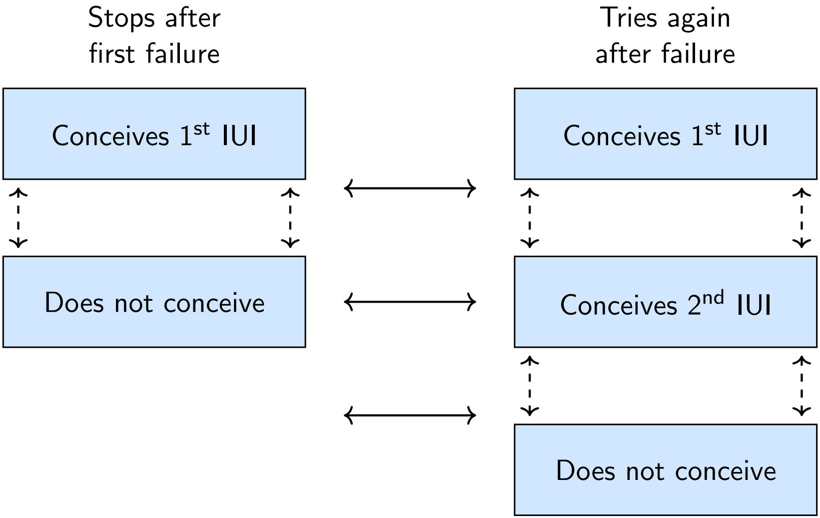
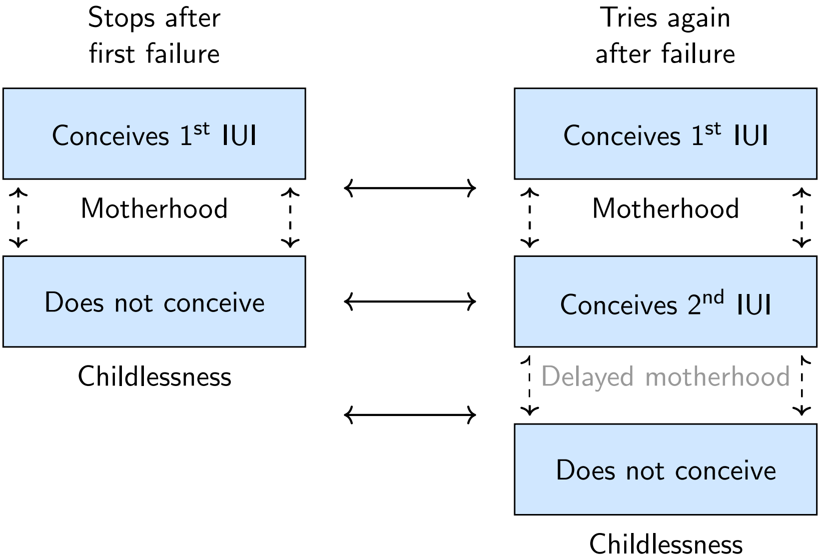


Simple World: Two Possible IUI Attempts and No Natural Conception



Simple World: What Is Observed

Stops after first failure

Tries again after failure

Conceive 1st IUI

Does not conceive

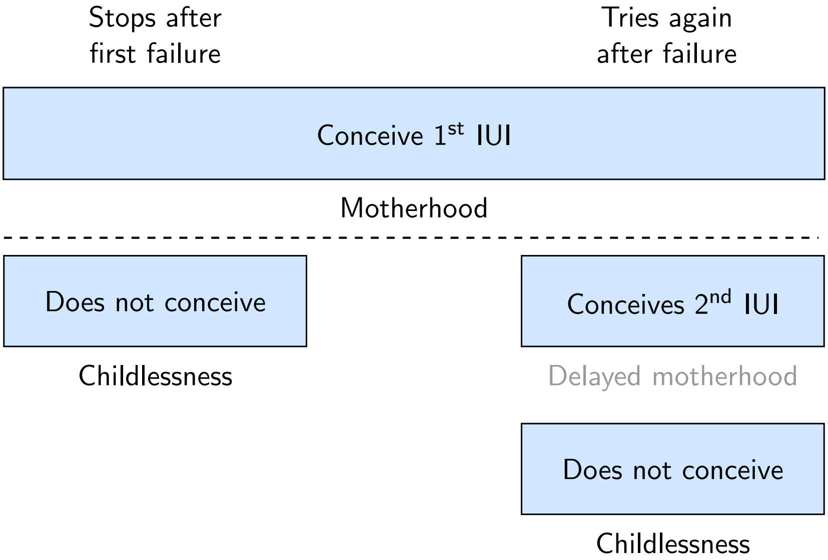
Conceives 2nd IUI

Does not conceive

Simple World: What Is Observed

Stops after first failure	Tries again after failure
Conceive 1st IUI	
Motherhood	
Does not conceive	Conceives 2nd IUI
Childlessness	Delayed motherhood
	Does not conceive
	Childlessness

Simple World: What Is Observed



Simple World: What Is Observed

Stops after first failure

Tries again after failure

Conceive 1st IUI

Motherhood

Does not conceive

Childlessness

Conceives 2nd IUI

Delayed motherhood

Does not conceive

Childlessness

% stops after fail =

Data and Outcomes

Sample:

- 15,500 cohabiting opposite-sex Dutch couples undergoing IUI for their first child

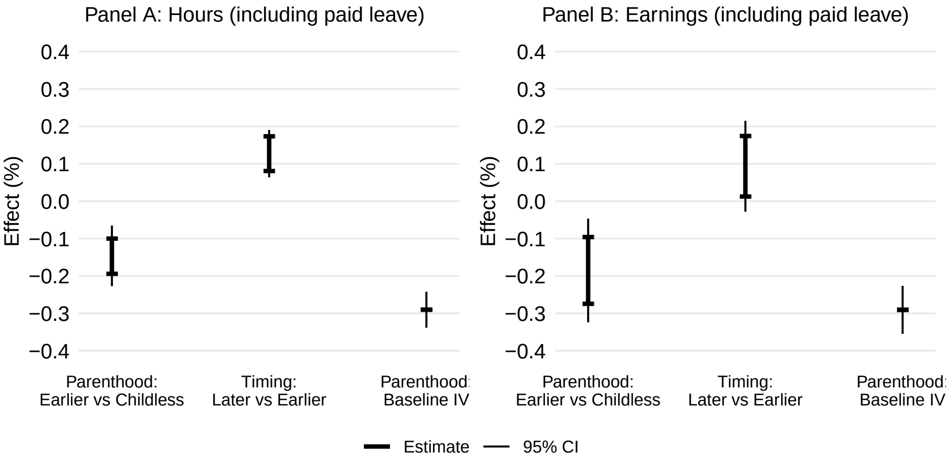
Main outcomes:

- Women's total earnings and work hours over the first seven years from first IUI

Main estimates:

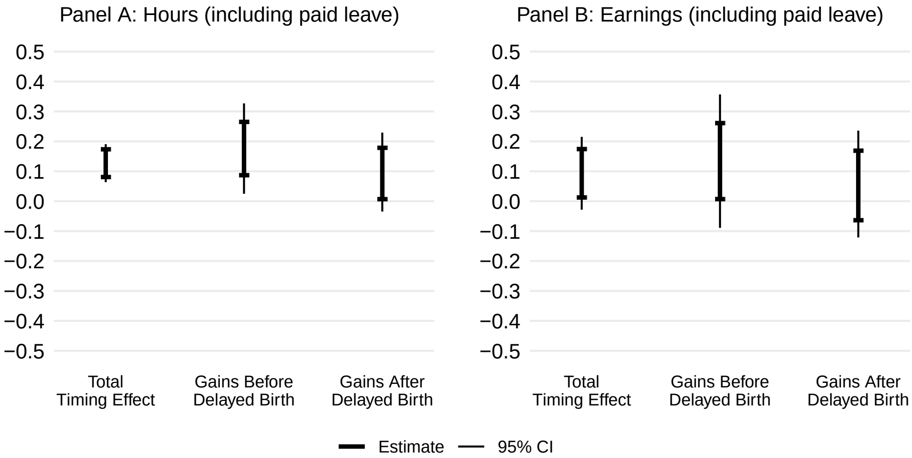
1. Parenthood effect: conception at first IUI vs remaining childless
2. Timing effect: conception after IUI sequence failure vs conception at first IUI
   - Average delay in conception $\sim$ 3 years
3. Naive IV parenthood effect estimates that assume no timing effects

Effect of Motherhood



- Later childbearing mitigates losses; ignoring it overstates the cost of motherhood
- No effects on men; parenthood causes 5%–50% of gender gaps.

Decomposing Timing Effect



- Benefits of delaying childbearing arise both before and after the delayed birth

Conclusion

Results:

- Motherhood causes up to half of gender inequality in hours and income
- Delaying reduces losses: a 5-year delay lowers hour (earnings) losses by 40% (20%)

External relevance:

- IUI is common; the sample matches the population on observables
- Many women would undergo IUI if natural conception failed
- Parenthood effects exceed Scandinavian IVF/contraceptive-failure estimates

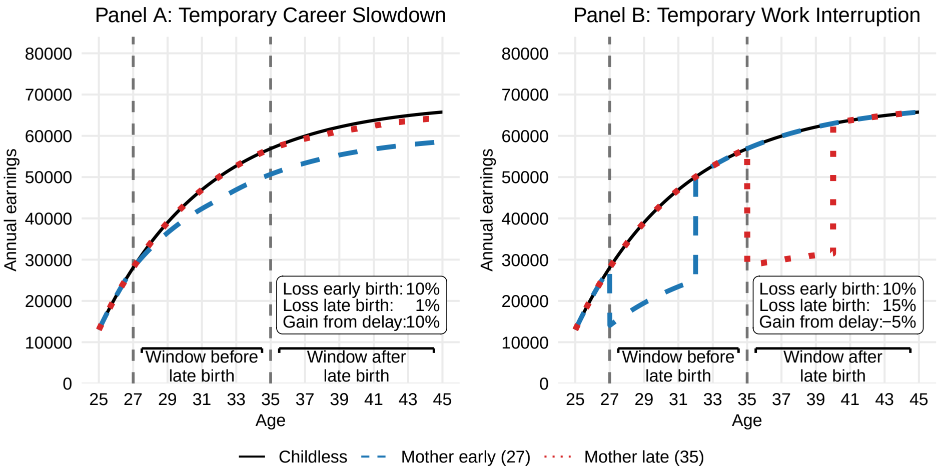
Policy:

- Enabling later childbearing may mitigate gender inequality
- Some gaps might persist even in the absence of children

Extensions

- **Selection into parenthood** Formal procedure Estimates
- Relation to partial identification literature Comparison
- Robustness Childless final period De-aging partners Mental health Bounds for non-depressed

Illustration of Timing Effects



How Existing Work Isolates Parenthood Effects

Per-period: effects may depend on current lifecycle stage and duration of motherhood:

$$\tau(\text{age}, \text{duration})$$

- Age = current career stage
- Duration = first child age
- **Age x Duration** = career stage at first birth

A) Standard IV approach (Lundborg et al., 2017, 2024):

$$\tau(\text{age}, \text{duration}) = \tau(\text{age})$$

B) Recursive IV approach (Bensnes et al., 2023; Gallen et al., 2023):

$$\tau(\text{age}, \text{duration}) = \tau(\text{duration}) \quad \text{(and homogeneous effects)}$$

- A) requires that duration does not matter, B) that career stage does not matter
- Both A) and B) require that **career stage at parenthood** does not matter
- Estimates differ substantially across approaches (Bensnes et al., 2023)

Life-cycle: Both A) and B) assume the timing effect offsets the parenthood effect.

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Bounding τ_{ATR}

Construct the moment:

$$m^L(G,\eta^0) = Y1_{\{Y<q(r(X_1),X_1)\}}\frac{Z_1}{e_1(X_1)} - Y(1-D)\prod_{j=1}^{A}\frac{(1-Z_j)}{(1-e_j(X_j))}$$

- G is the observed data vector
- η^0 contains the following:
  - $e_j(X_j) = \Pr(Z_j = 1|X_j)$
  - $q(r(X_1), X_1)$ is the $r(X_1)$-th quantile of Y given X_1 and $Z_1 = 1$
  - $r(X_1)$ identifies the covariate-conditional relier share

[Conditional Sequential Unconfoundedness] $(Y(k), R, W) \perp\!\!\!\perp Z_j \mid X_j$ for all j, k, and $X_j, A \geq j$.

Theorem (Lower Bound)

Under conditional sequential unconfoundedness and regularity, the sharp lower bound on τ_{ATR} is $\mathbb{E}[m^L(G,\eta^0)]/\,\mathbb{E}[r(X_1)]$.

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How Much Does Delayed Childbearing Mitigate Career Costs?

Focus on midpoints of bounds

- Necessarily suggestive; comparing short run estimates for reliers and non-reliers

Before delayed birth:

- Absence of children: +16% annual hours, +19% earnings

After delayed birth:

- Gains shrink to +5% hours, +2% earnings

Lifetime implications (33-year horizon from age 32)

- 1-year delay $\Rightarrow$ +5.5% hours, +2.5% earnings
- 5-year delay $\Rightarrow$ +7% hours, +4.5% earnings

Baseline long-run parenthood losses: –15% hours, –20% earnings

Five-year delay reduces losses to: –9% hours, –16% earnings

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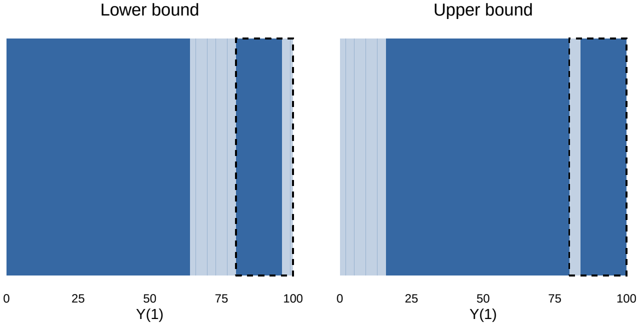
Intuition: Motherhood Outcome $Y_1(1)$—Covariates

Pre-IUI covariates can help narrow the bounds:

- Can identify relier share at each covariate value
- Baseline bounds assume extreme scenarios where reliers have highest or lowest treated outcomes
- These distributions of treated outcomes might be inconsistent with conditional relier shares



Estimating the Bounds

Distribution of $m^L(G, \eta^0)$ is complicated by $q(r(X_1), X_1)$

- Semenova (2023) addresses a closely related inference challenge
- Double/debiased machine learning approach
  1. Adjust $m^L(G, \eta^0)$ to make it insensitive to small error in $q(r(X_1), X_1)$
  2. Sample splitting
- Asymptotic inference as if $q(r(X_1), X_1)$ was known

New moment:

$$\psi^L(G, \xi^0) = m^L(G, \eta^0) + corr(G, \xi^0)$$

Identifies same parameter:

$$\mathbb{E}[\psi^L(G, \xi^0)] = \mathbb{E}[m^L(G, \eta^0)]$$

Insensitive to estimation error in $q(r(X_1), X_1)$:

$$\partial_{q(.)} \mathbb{E}[\psi^{L+}(G, \xi_r)|X_1]|_{\xi_r = \xi_r^0} = 0 \; a.s.$$

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Assisted Conception Procedures

- IUI (main procedure): sperm injected into uterus
  - Minimally invasive, primary ACP in most countries
  - “Free” in NL
- IVF (secondary procedure): embryo inserted into uterus
  - Invasive treatment, performed under sedation/anesthesia
  - Eggs retrieved through the vaginal wall using a specialized needle
  - In NL, first 3 free; each subsequent costs between 1000 and 4000 EUR

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Institutions

- Dutch family friendly polices similar to OECD average
  - 16 weeks of fully paid pregnancy+maternity leave
  - 1 week of paternity leave
  - Average time in child care similar to OECD average
  - Net child care cost 10% median household income
- Dutch employment intensity similar to OECD average
  - Employment among parents and non-parents relatively high
  - Part time work much more common
  - Approximately 15% two-parent families have both partners working part-time

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Data

Administrative data from Statistics Netherlands

- Comprehensive hospital records cover fertility treatments from 2012 to 2017: procedure date and type
  - Success imputed as having child born within 10 months
- Tax records cover work hours and income from 2011 to 2023
  - Include maternity leave and pay
  - Main bounds account for uncertainty around actual work hours
- Birth dates, legal family connections, cohabitation
- Dispensed medication registry

Main sample: cohabiting opposite-sex couples undergoing IUI for their first child between 2013 and 2016: 15,523

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Balance in 1^{st} IUI

Back (data) Back (representativeness)

Table 1: First IUI Outcomes and Descriptives

	Success (1)	Fail (2)	Dif. (1)-(2)	IPW dif. (1)-(2) cond.	Rep. (5)	Suc. vs rep. (1)-(5)
Work (W)	0.912	0.916	-0.004	-0.009	0.936	-0.024
	[0.283]	[0.277]	(0.008)	(0.008)	[0.244]	(0.007)
Work (P)	0.894	0.885	0.009	0.002	0.897	-0.002
	[0.307]	[0.319]	(0.009)	(0.009)	[0.304]	(0.008)
Hours (W)	1300.012	1298.876	1.136	-1.951	1310.923	-10.911
	[547.832]	[558.316]	(15.730)	(16.119)	[544.468]	(14.554)
Hours (P)	1513.337	1494.541	18.796	3.345	1497.603	15.734
	[635.121]	[656.050]	(18.457)	(19.041)	[651.043]	(17.403)
Earn. 1000s EUR (W)	29.358	29.648	-0.290	0.203	26.555	2.803
	[18.000]	[18.911]	(0.531)	(0.561)	[15.989]	(0.427)
Earn. 1000s EUR (P)	38.082	38.060	0.022	0.322	33.862	4.220
	[25.425]	[26.525]	(0.745)	(0.774)	[24.148]	(0.646)
Bachelor deg. (W)	0.512	0.494	0.018		0.518	-0.007
	[0.500]	[0.500]	(0.014)		[0.500]	(0.013)
Bachelor deg. (P)	0.425	0.410	0.014		0.430	-0.005
	[0.494]	[0.492]	(0.014)		[0.495]	(0.013)
Age (W)	31.373	32.060	-0.687		28.840	2.533
	[3.889]	[4.265]	(0.119)		[3.896]	(0.104)
Age (P)	34.088	34.856	-0.768		31.415	2.673
	[4.968]	[5.500]	(0.154)		[4.803]	(0.128)
Observations	1,411	11,323			171,180	
Joint p-val.			0.001	0.955		0.000

Note: *Success* – average among women whose first IUI succeeded; *Fail* – average among women whose first IUI failed; *Dif.* – difference between *Success* and *Fail*; *IPW dif.* – difference adjusted for age and education using inverse probability weights from the baseline specification; *Rep.* – average in representative sample of women who conceived their first child without assisted conception procedures; *Suc. vs rep* – difference between *Success* and *Rep.*. Reference year: year of first IUI (IUI sample); 9 months before first birth (representative sample). IUI sample: women who underwent intrauterine insemination for their first child between 2013 and 2016, with no prior assisted conception procedures, cohabiting with a male partner in the year prior to the reference year. Representative sample: women with no assisted conception procedures before first birth, cohabiting with a male partner in the year prior to the reference year, with reference year between 2013 and 2016. Labor market outcomes measured in the year before the reference year; age measured in the reference year. *Bachelor deg.* – indicator for completing a bachelor's degree. *Earn.* – earnings, (W) – woman, (P) – partner. Standard deviations in brackets. Standard errors in parentheses.

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Balance in Subsequent ACPs

Table 2: Balance in Later ACPs

	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7	Z_8	Z_9	Z_{10}
Work (W)	0.009	-0.004	0.022	0.014	0.039	-0.003	-0.011	0.022	0.030
	(0.010)	(0.011)	(0.011)	(0.012)	(0.012)	(0.017)	(0.018)	(0.019)	(0.024)
Work (P)	0.006	0.016	0.012	0.020	-0.004	-0.004	-0.019	0.017	0.030
	(0.010)	(0.010)	(0.012)	(0.012)	(0.015)	(0.015)	(0.019)	(0.020)	(0.027)
Hours (W)	32.885	-4.482	52.999	41.332	81.957	11.894	-18.836	72.659	24.819
	(18.721)	(20.032)	(21.045)	(22.686)	(25.131)	(31.187)	(32.937)	(38.210)	(48.490)
Hours (P)	21.655	24.730	23.756	38.965	9.666	-6.580	-28.458	30.525	43.722
	(21.018)	(21.089)	(23.574)	(25.255)	(30.585)	(31.513)	(37.976)	(44.856)	(52.821)
Income 1000s € (W)	1.481	-0.015	1.685	1.802	2.086	0.150	-0.043	0.866	-0.444
	(0.615)	(0.624)	(0.767)	(0.830)	(0.913)	(1.000)	(1.092)	(1.234)	(1.629)
Income 1000s € (P)	-0.749	1.002	2.040	0.800	0.774	0.025	0.259	-0.324	0.149
	(0.835)	(0.912)	(1.066)	(1.115)	(1.424)	(1.424)	(1.563)	(1.737)	(2.203)
Observations	12,974	10,774	8,726	6,977	5,411	3,944	2,723	1,850	1,174
Joint *p*-val.	0.175	0.976	0.234	0.303	0.140	1.000	0.956	0.704	0.917

Note: Each column describes the difference in average characteristics between women for whom the respective ACP succeeds and those for whom it fails, among those who undergo the procedure, using inverse probability weights for each ACP following the main specification. Labor market outcomes and age measured year before first treatment. (W) - woman, (P) - partner. Standard errors in parentheses.

Estimated Success Probabilities

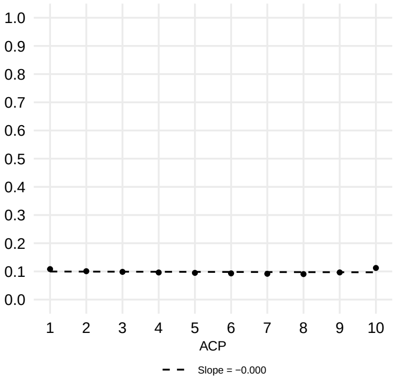

Figure 1: Estimated Success Probabilities

Comparison with Lee (2009)

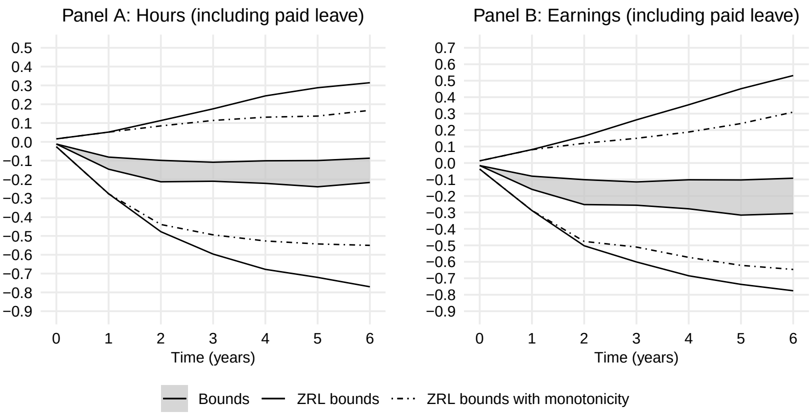

Figure 2: Comparison with Lee (2009) Bounds for Effects on Women

Quantifying Selection Using the Event Study Approach

Kleven et al. (2024) compare mothers t years after childbirth to women one year before

- Most gender inequality is explained by these differences between women
- Event study estimates may be biased due to selective timing
- But differences from IV/bounds need not imply bias, even in the same sample

1. I proxy fertility timing using the timing of first IUI
2. Compare the actual trajectories of women who attempted conception but remained childless with those imputed from women who are about to attempt conception but ultimately remain childless

- **Event study comparison of women with different timing absent children**
- **Same population as bounds: selection vs causal effects directly comparable**

Back (extensions) Conclusion

Placebo Event Study

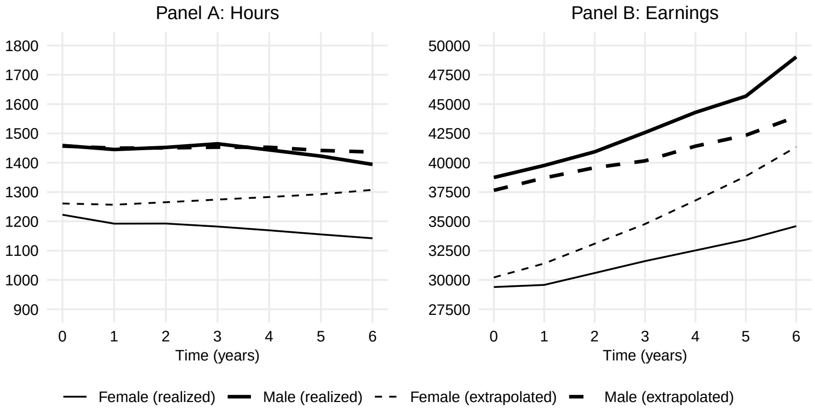

Figure 3: Placebo Event Study

Back (extensions) Conclusion

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Gender Inequality: Causal vs Selection

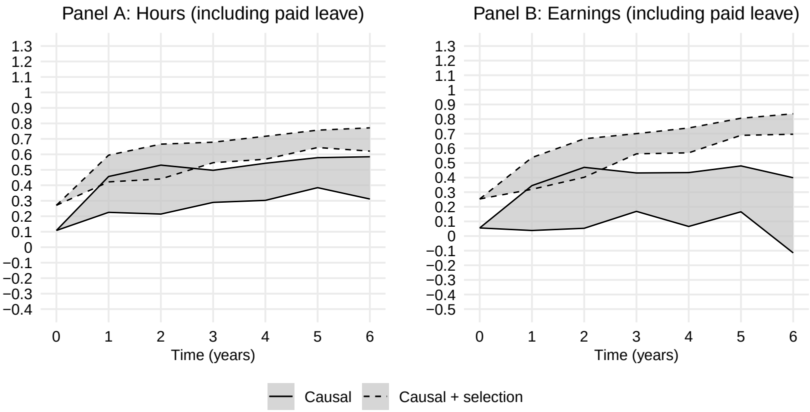

Figure 4: Share of Gender Inequality Explained by Selection and Parenthood

Back (extensions) Conclusion
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Mental Health and Assisted Conception

Mental health consequences associated with failure to conceive are a part of the story:

- Unmet fertility goals may negatively impact mental health, and in turn, labor market outcomes

There are, however, additional concerns:

- Mental health issues caused specifically by failed conception or ACPs (external)
  - Focusing on artificial insemination helps mitigate this
- Large impacts unique to ACP families (external)
- Worsened mental health by threatening monotonicity (internal)

In practice, these impacts are likely small (Lundborg et al., 2024)

Antidepressant uptake | Back (extensions) | Conclusion

ToC

Monotone Bounds for Non-depressed Childless Women

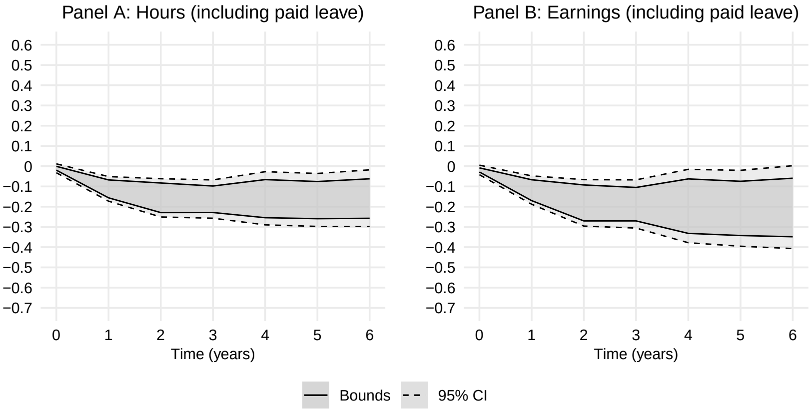

Figure 5: Monotone Bounds for Women Who Would Not Uptake Antidepressants if They Were to Remain Childless

Back (extensions) Conclusion

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Effect of IUI Failure on Antidepressant Use

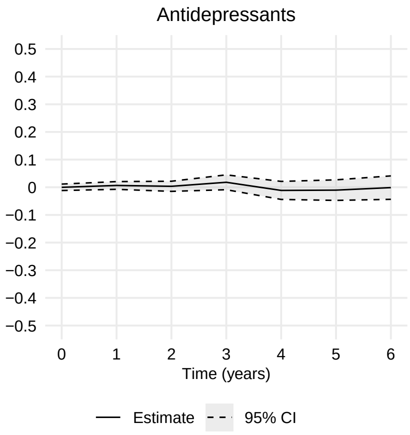

Figure 6: Estimates for effect on antidepressant take-up

Back (extensions) Conclusion
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Monotone Bounds: Women Who Remain Childless

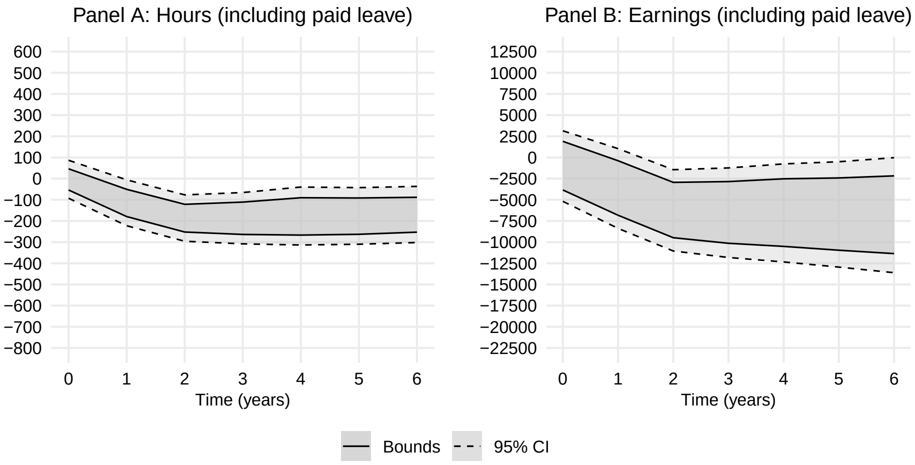

Figure 7: Monotone Bounds Using Completed Fertility

Parenthood Effect Bounds: Correcting for Partner's age

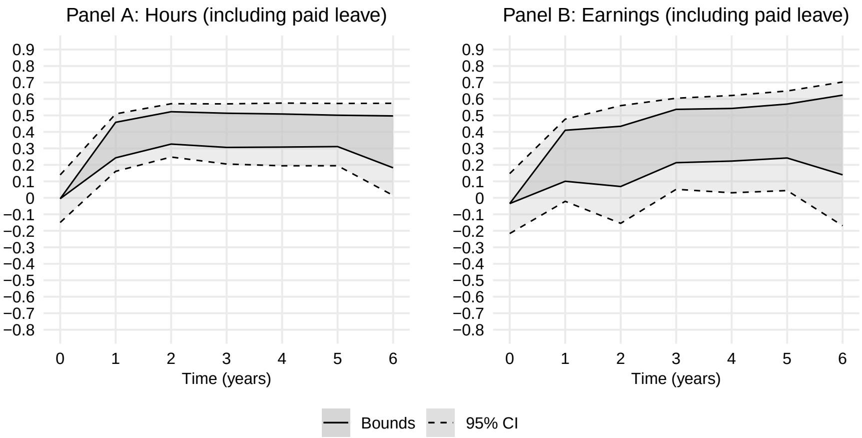

Figure 8: Bounds Measuring Male Income at Same Age as Female

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